

Mental Health Prediction & Risk Assessment – CRISP-DM Report.

Introduction.

This project delivers an integrated mental health prediction and risk assessment tool, combining **natural language processing** for symptom-based disorder identification with **machine learning** for lifestyle-based risk evaluation. The system is designed for use by individuals, clinicians, and mental health practitioners, providing early detection, risk scoring, and actionable recommendations.

The platform supports two complementary functionalities:

1. **Signs-Based Disorder Prediction** – Users provide textual descriptions of symptoms, which are transformed into vector embeddings and classified into likely mental disorders. The model then returns relevant **self-care** and **professional** recommendations.
2. **Lifestyle-Based Risk Assessment** – Users input structured lifestyle and demographic data, and the model predicts both the likelihood and severity of mental health risk.

By combining these approaches, the application aims to provide **holistic mental health evaluation**, increasing awareness, promoting early intervention, and supporting informed decision-making.

1. Business Understanding

Project Goal:

Develop a user-facing mental health application with two integrated capabilities:

1. **Signs-Based Disorder Prediction** – The user inputs signs/symptoms in free-text form, and the system predicts the most likely mental disorder(s).
2. **Lifestyle-Based Risk Assessment** – The user provides structured lifestyle and demographic features, and the system evaluates the risk of developing a mental health disorder and its severity.

Primary Stakeholders:

- Clinicians and mental health practitioners
- Users seeking early screening

- Developers and maintainers of the UI system

Success Criteria:

- High prediction accuracy (precision/recall) for disorders
- Reliable and interpretable risk scores for lifestyle-based predictions
- User-friendly, interactive UI with clear recommendations

2. Data Understanding

This project integrates **two distinct datasets**:

1. Lifestyle & Demographics Dataset (for Risk Classifier)

- **Rows:** 50,000
- **Columns:** 15
- **Features:**
 - Age
 - Gender
 - Occupation
 - Mental_Health_Condition (Target – Binary)
 - Severity
 - Consultation_History
 - Stress_Level
 - Sleep_Hours
 - Work_Hours
 - Physical_Activity_Hours
 - Social_Media_Usage
 - Diet_Quality
 - Smoking_Habit
 - Alcohol_Consumption
 - Medication_Usage
- **Purpose:** Predict risk of mental health disorder and classify severity levels.

2. Signs & Recommendations Dataset (for Disorder Identifier)

- Contains textual signs/symptoms for various mental disorders.
- Each disorder's signs are merged into a single aggregated text entry for training.
- **Output:**
 - Predicted disorder name(s)
 - Associated recommendations:
 - **Self** – Lifestyle and coping strategies
 - **Professional** – Medical and therapeutic advice

Preprocessing & Encoders Used:

- Label encoding for categorical features (Gender, Occupation, etc.).
- SentenceTransformer embeddings for textual symptom data.

Other Models Used:

CatBoostClassifier, GradientBoosting, LogisticRegression, RandomForest.

3. Data Preparation

Steps for Lifestyle Data:

- Label encoding categorical variables.
- Scaling/normalization of numeric variables where appropriate.
- Train/test split for evaluation.
- Handling missing values (imputation if necessary).

Steps for Signs Data:

- Text cleaning (lowercasing, punctuation removal if applied).
- Aggregation of all symptoms for each disorder into a single text sample.
- Embedding generation via SentenceTransformer (all-MiniLM-L6-v2).
- Alignment of symptom entries with corresponding recommendations.

4. Modeling

Signs-Based Model:

- **Algorithm(s) Used:** Gradient Boosting, CatBoostClassifier.
- **Input:** Symptom embeddings.
- **Output:** Predicted mental disorder class.

Lifestyle-Based Model:

- **Algorithm(s) Used:** LogisticRegression, RandomForest, GradientBoosting.
- **Input:** Encoded categorical + numeric lifestyle features.
- **Output:** Binary classification (Disorder Risk: Yes/No) + Severity prediction.

5. Evaluation

Metrics Used:

- Accuracy Score
- Precision, Recall, F1-score (via classification report)
- ROC AUC Score
- Confusion Matrix

Signs-Based Model:

- Evaluated using multi-class metrics per disorder.
- May use one-vs-rest ROC AUC for each disorder.

Lifestyle-Based Model:

- Evaluated for binary classification and severity stratification.
- Threshold tuning for “Low/Medium/High” risk categories.

6. Deployment

Planned/Implemented Steps:

- Streamlit-based interactive UI with two tabs:
 - Tab 1: Symptom input → disorder prediction + recommendations.
 - Tab 2: Lifestyle input → risk score + severity.
- Option to build FastAPI backend for scalable inference.
- Model persistence via Pickle/Joblib.

Operational Considerations:

- Include medical disclaimers in the UI.
- Implement monitoring for model drift.
- Allow regular model retraining when new data becomes available.